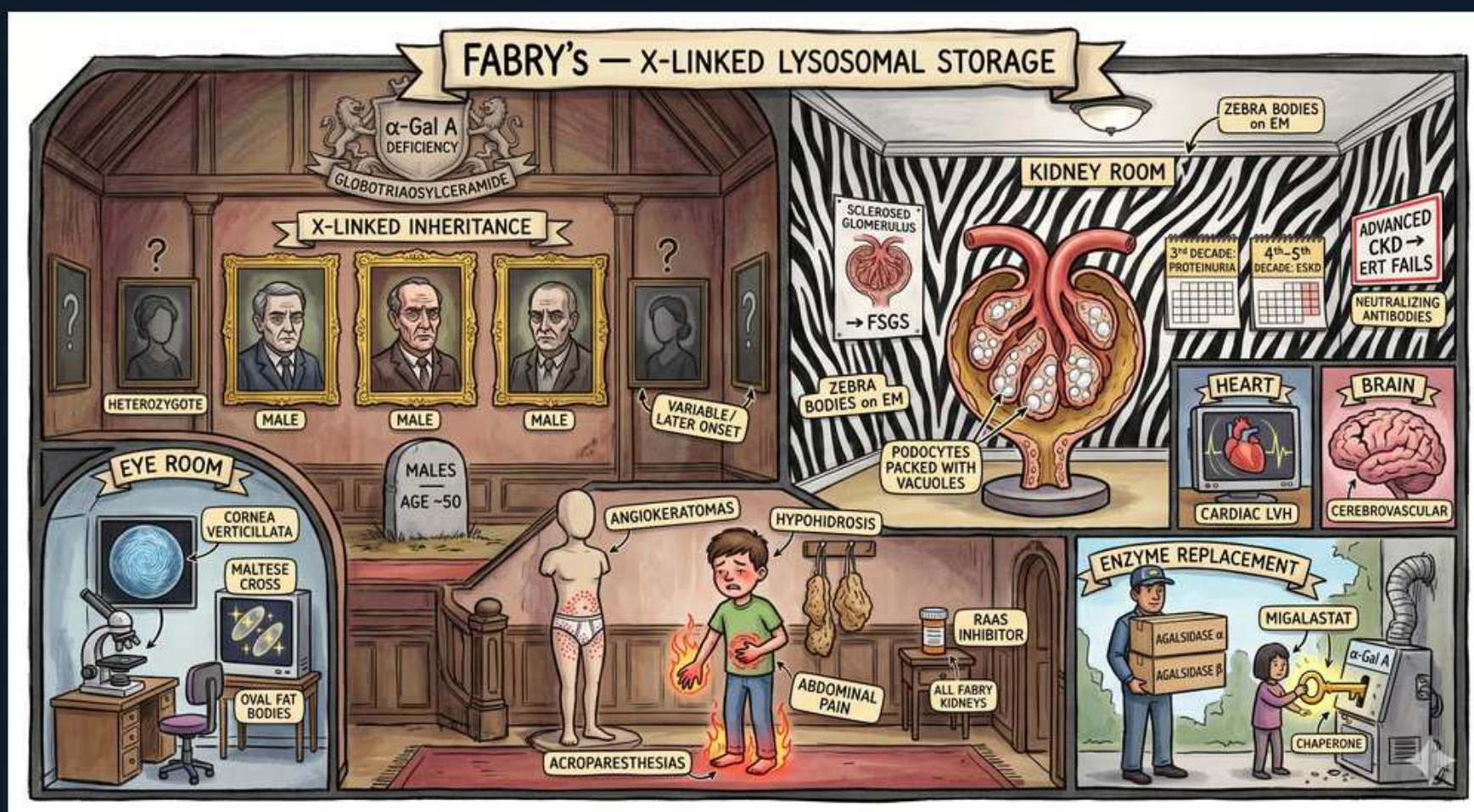


Fabry Disease — X-Linked Lysosomal Storage Nephropathy



1. α-Gal A Deficiency

WHAT YOU SEE

Plaque, top-left wall

WHAT IT ENCODES

Fabry disease is caused by deficiency of the lysosomal enzyme alpha-galactosidase A (GLA gene). Substrate globotriaosylceramide (Gb3/GL-3) accumulates in lysosomes of endothelium, podocytes, cardiomyocytes, neurons, and cornea. It is the second most common lysosomal storage disease.

BOARD TRAP

Don't confuse with alpha-glucosidase deficiency (Pompe) — Fabry is alpha-GALACTOSIDASE A and accumulates Gb3, not glycogen.

2. X-Linked Inheritance

WHAT YOU SEE

Portrait gallery of males

WHAT IT ENCODES

GLA is on the X chromosome (Xq22). Hemizygous males are fully affected; heterozygous females range from asymptomatic to fully symptomatic due to random X-inactivation (lyonization). No male-to-male transmission.

BOARD TRAP

Females are NOT obligate carriers-only — random X-inactivation means heterozygous women can have significant disease including renal/cardiac involvement.

3. Cornea Verticillata

WHAT YOU SEE

Eye room, slit-lamp whorls

WHAT IT ENCODES

Whorl-like corneal opacities (cornea verticillata) seen on slit-lamp exam; a near-universal, asymptomatic sign that doesn't affect vision. Useful early clinical clue and helps identify carrier females.

BOARD TRAP

Cornea verticillata can also be drug-induced (amiodarone, chloroquine, hydroxychloroquine) — context distinguishes Fabry from drug deposits.

4. Maltese Cross / Oval Fat Bodies

WHAT YOU SEE

Microscope, lipid bodies

WHAT IT ENCODES

Urinary sediment shows oval fat bodies and 'Maltese cross' birefringent lipid under polarized light, reflecting lipiduria from glomerular involvement. Also seen in any heavy nephrotic-range proteinuria.

BOARD TRAP

Maltese crosses indicate lipiduria/nephrotic-range proteinuria generally — not specific to Fabry; pair with systemic clues.

5. Angiokeratomas

WHAT YOU SEE

Skin lesions on torso

WHAT IT ENCODES

Clusters of dark red-purple angiokeratomas, classically in a 'bathing-trunk' distribution (groin, buttocks, umbilicus). Result from Gb3 accumulation in dermal vessels. Early, visible diagnostic clue.

BOARD TRAP

Bathing-trunk angiokeratomas + neuropathic pain in a young man point to Fabry, not benign cherry angiomas or vasculitic purpura.

6. Acroparesthesias

WHAT YOU SEE

Burning hands/feet, flames

WHAT IT ENCODES

Episodic burning pain in hands and feet (acroparesthesias / Fabry crises), often the earliest childhood symptom, triggered by heat, exercise, or fever. Due to small-fiber neuropathy from Gb3 in dorsal root ganglia.

BOARD TRAP

Burning extremity pain in a child is easily misread as 'growing pains' or psychogenic — the Fabry clue is the triggered crises plus family history.

7. Hypohidrosis

WHAT YOU SEE

Label center, no sweat

WHAT IT ENCODES

Decreased sweating (hypohidrosis/anhidrosis) from autonomic and sweat-gland involvement causes heat intolerance and contributes to exercise-triggered pain crises. Anhidrosis is more typical than hyperhidrosis.

BOARD TRAP

Fabry causes HYPOhidrosis (heat intolerance), not excessive sweating — a discriminator from autonomic hyperactivity states.

8. FSGS / Sclerosed Glomeruli

WHAT YOU SEE

Kidney room, glomerulus diagram

WHAT IT ENCODES

Glomerular involvement produces proteinuria, secondary FSGS, and progressive global glomerulosclerosis. Proteinuria typically appears in the 2nd–3rd decade and progresses to ESRD by the 4th–5th decade if untreated.

BOARD TRAP

The FSGS here is secondary to storage — primary podocytopathy steroids won't help; the answer is enzyme replacement plus RAAS blockade.

9. Podocytes Packed with Vacuoles

WHAT YOU SEE

Glomerulus center, vacuolated cells

WHAT IT ENCODES

Podocytes show prominent cytoplasmic vacuolization from lysosomal Gb3 storage on light microscopy — a key histologic clue. Foam cells/vacuolated tubular and endothelial cells are also seen.

BOARD TRAP

Vacuolated/foamy podocytes mimic the appearance after some drugs (e.g., gentamicin, chloroquine 'phospholipidosis') — EM zebra bodies confirm Fabry.

10. Zebra Bodies on EM

WHAT YOU SEE

Top banner, zebra-stripe wall

WHAT IT ENCODES

Electron microscopy reveals lamellated 'zebra bodies' / myeloid bodies — concentric lamellar inclusions of stored glycosphingolipid within lysosomes. This is the pathognomonic ultrastructural finding.

BOARD TRAP

Zebra/myeloid bodies are the EM hallmark; chloroquine/amiodarone can cause similar lamellar bodies, so correlate with enzyme assay and GLA mutation.

11. Cardiac LVH

WHAT YOU SEE

Heart room icon

WHAT IT ENCODES

Gb3 in cardiomyocytes causes concentric left ventricular hypertrophy, conduction abnormalities, and arrhythmia; cardiac disease is a leading cause of death. A cause of unexplained LVH / HFpEF in adults.

BOARD TRAP

In unexplained LVH, Fabry is on the differential with HCM and amyloidosis — short PR interval and angiokeratomas/neuropathy steer toward Fabry.

12. Cerebrovascular / Stroke

WHAT YOU SEE

Brain room icon

WHAT IT ENCODES

Premature TIA/stroke (often posterior circulation) from Gb3 vasculopathy and endothelial dysfunction. Consider Fabry in young patients with cryptogenic stroke, especially with renal/cardiac signs.

BOARD TRAP

Cryptogenic stroke in a young patient should prompt Fabry screening — don't anchor only on PFO or thrombophilia.

13. RAAS Inhibitor

WHAT YOU SEE

Pill bottle, lower center

WHAT IT ENCODES

ACE inhibitors/ARBs reduce proteinuria and slow progression of Fabry nephropathy, used alongside disease-specific therapy. Standard adjunct for any proteinuric CKD.

BOARD TRAP

RAAS blockade is supportive antiproteinuric therapy — it does not replace the missing enzyme or clear Gb3.

14. Enzyme Replacement (Agalsidase)

WHAT YOU SEE

Banner, agalsidase boxes

WHAT IT ENCODES

Recombinant agalsidase alfa/beta is the mainstay disease-modifying therapy, replacing alpha-Gal A to clear Gb3; earlier initiation gives better organ outcomes. Neutralizing antibodies can reduce efficacy (depicted on the calendar wall).

BOARD TRAP

ERT slows progression but is most effective started EARLY before advanced fibrosis — once ESRD/severe LVH develops, benefit drops.

15. Migalastat (Chaperone)

WHAT YOU SEE

Migalastat label, far right

WHAT IT ENCODES

Migalastat is an oral pharmacologic chaperone that stabilizes amenable (mutation-specific) misfolded alpha-Gal A, restoring residual activity. Only works for patients with 'amenable' GLA mutations.

BOARD TRAP

Migalastat is NOT for every patient — only amenable missense mutations respond; nonsense/large deletions still need ERT.